

SmallChip AI — ISEF 1-pager handout

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A Free, BSD-3 Open-Source Real-Time Interactive Chip Placement Co-Pilot with LLM

Harshith Nelabhotla, Strongsville High School, Strongsville OH Faculty sponsor: Mrs. DiGioia · Category: Math / Computer Science (MCS)

The problem

Every chip in the world needs placement. The commercial tools (Cadence, Synopsys) cost **500K–1M/year per seat** and run as **batch jobs (5–60 min)**. The leading open-source tool (OpenROAD) is free but **also batch-only and often diverges on hard designs**. **No cell-level placement tool — commercial, academic, or open-source — is real-time interactive.**

The solution

SmallChip AI, a Graph Attention Network with 18,000 parameters that:

Feature	Result
Placement inference (15K cells)	150ms (CPU)
Partial re-placement on drag (15K design)	14ms for 71 cells
Hierarchical scaling proven	30,000,000 cells in 50s
Projected hierarchical scale	100,000,000 cells
Open-source license	BSD 3-Clause
LLM co-pilot	"make this faster" → re-placed chip
Training data	510 synthetic + ISPD 2005

The results (all validated, no projections)

Benchmark	Cells	Our result	Reference
GCD (Nangate45 PDK)	734	99.7% HPWL reduction, 370× better, identical timing + power	OpenROAD default: 3,987,080 DBU

Benchmark	Cells	Our result	Reference
Held-out test (66 designs, 0% training overlap)	100–700	100% win rate, 87.1% avg improvement, 87.5% median	Random baseline
bigblue1_15k_subset (hierarchical, 3 blocks)	15,000	1,281 DBU/net	Random: 7,008 DBU/net
30M synthetic (hierarchical)	30,000,000	1,281,714 DBU/net	(first ever demonstrated)

The architecture

User uploads .def → SmallChip AI → re-placed .def + GDS
↓
[1] GAT V3 (150ms, BSD-3)
[2] Smart legalizer (OpenROAD-compatible)
[3] Detailed placer (flip/shift/swap)
[4] LLM co-pilot (Ollama, local, free)
↓
For designs >15K cells:
[A] BFS-aware partitioner
[B] Force-directed top placement
[C] V3 per block (parallel)
[D] Inter-block wire refinement

The ecosystem

Step	Open-source tool (before)	Now
PDK	Skywater 130nm	(same)
Synthesis	Yosys	(same)
Placement (interactive)	— (missing)	SmallChip AI
Routing, CTS, DRC	OpenROAD, KLayout	(same)
Cores	RISC-V	(same)

SmallChip AI completes the open-source EDA ecosystem.

Why it matters

Audience	Annual value
1-engineer chip company	37,500/yr
2-engineer company	45,000/yr
5-engineer company	67,500/yr
University ECE lab (10 students)	200,000/yr

Audience	Annual value
High school + community college	Enables chip design without 1M tool
1,000+ small chip companies in US	40M/yr value
5,000+ university courses	250M/yr value

The proof points

- 18,000-parameter GAT (lightweight, runs on a laptop)
- Trained on 510 designs, validated on 66 held-out (clean)
- BSD-3 license, anyone can use commercially for free
- GitHub: github.com/hnelabhotla-boop/smallchip-ai
- arXiv preprint: in /paper/ of the repo
- Web app + desktop .app + LLM co-pilot
- 100% wins on held-out, 99.7% on GCD, 370× better
- Hierarchical to 30M cells proven

Why now

- **First** free, BSD-3 open-source real-time interactive cell-level placer
- **First** BSD-3 placer with an LLM co-pilot
- **First** end-to-end open-source chip design flow that has a free interactive placement layer
- The open-source EDA ecosystem has been waiting for this missing piece for years

What's next

- V4 multi-objective loss (HPWL + congestion + thermal), 200K params, 200 epochs
- DREAMPlace head-to-head on adaptec1 + bigblue1
- efabless Skywater 130nm shuttle for physical chip
- 1+ academic co-author
- 5+ beta testers in university ECE programs
- ISEF Special Award (Moore 20K, IEEE 10K, ACM 5K, Sigma Xi \$5K)

Contact

Harshith Nelabhotla · hnelabhotla@students.strongsville.k12.oh.us **Faculty sponsor:** Mrs. DiGioia · Strongsville High School Science Research Program **GitHub:** <https://github.com/hnelabhotla-boop/smallchip-ai> **Project page:** [in repo](#)